

Intelligence is Physical: Energy, Information, & a Thermodynamically Grounded Theory of Minds & Machines

Abstract: Despite the prominence of formalist theories of intelligence within the artificial intelligence research tradition, intelligence cannot be fully understood as a purely formal or symbolic process. Instead, it emerges from the physical interactions of energy, matter, and information across nested levels of complexity, from basic physical structures to life, neural systems, and adaptive behaviour. In this paper, we introduce a hierarchical, thermodynamically grounded theory of intelligence that emphasizes the coupling of sensory, cognitive, and motor processes with metabolic and informational constraints. By explicitly considering the energy costs and physical instantiation of computation, our theory provides a principled framework for understanding natural intelligence and guiding the design of sustainable, efficient, and adaptive artificial systems.

Keywords: Thermodynamic; intelligence; nested ontology; coupling logic; scaling laws; information engine; entropy

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§1 Introduction

‘To train the large language model (LLM) powering Chat GPT-3, for example, almost 1,300MW-h of energy was used, according to an estimate by researchers from Google and the University of California, Berkeley, a similar quantity of energy to what is used by 130 American homes in one year. Furthermore, an analysis by OpenAI suggests the amount of power needed to train AI models has been growing exponentially since 2012, doubling roughly every 3.4 months as the models become larger and more sophisticated. However, our energy production capacity is not increasing as steeply, and actually doing so is likely to further contribute to global warming: generating electricity is the single biggest contributor to climate change given that coal, oil, and gas are still widely used to generate electricity, compared to cleaner energy sources.’ —Sandrine Ceurstemont (2025, p. 15)

‘We have got on to slippery ice where there is no friction and so in a certain sense the conditions are ideal, but also, just because of that, we are unable to walk. We want to walk: so we need friction. Back to the rough ground!’ —Ludwig Wittgenstein (1953, § 107)

The artificial intelligence (hereafter: AI) research tradition is inundated with formalist theories of intelligence. They include:

(T1 or the Church-Turing thesis) For any real-world problem that we believe to be computable, we should, in principle, be able to construct a simple abstract machine (known as a Turing machine) that computes it (Church, 1936; Turing, 1936);

(T2 or the McCulloch-Pitts model of the neuron) The brain can, in principle, be modelled as a network of logical elements (known as artificial neurons) that receive input signals, perform weighted sums, and fire output signals if a threshold is satisfied (McCulloch and Pitts, 1943);

(T3 or information theory) Communication can, in principle, be modelled as a transmission of a message from a source to a receiver, with information being measured in terms of how much uncertainty is reduced when a message is received (Shannon, 1948).

T1 implies that all forms of reasoning or problem-solving, as long as they follow definite rules, can be reduced to symbol manipulation. T1 treats thought as an abstract process of computation, shorn of its roots in the physical world. T2 implies that networks of logical elements can, in principle, perform logical calculus, paving the way for artificial neural networks. However, T2 ignores the physical, biological, chemical, and evolutionary realities of biological neurons, reducing them to logical gates in a circuit. T3 makes communication and computing mathematically tractable, yet it reduces information to a purely formal concept, decoupled from matter, energy, and semantics.

More generally, T1–T3 are top-down theories that decouple information gradients from matter and energy gradients, thereby obscuring, ignoring,

or abstracting from the embodied, metabolic, and evolutionary constraints that make cognition possible. At least some AI researchers have sought to foreground the physical nature of intelligence:

(T4 or the physical symbol system hypothesis) A physical symbol system has the necessary and sufficient means for general intelligent behaviour (Newel and Simon, 1976);

(T5 or the physical grounding hypothesis/embodiment) Intelligence should be grounded in an interaction between a system and its surroundings (Anderson, 2003; Brooks, 1990);

(T6 or the embodied intelligence hypothesis) Intelligence is confined to the brain, but partly emerges from the interaction of the physical body with the environment and the task (Chrisley, 2003; Pfeifer, Lungarella, and Iida, 2007).

Although T4, T5 and T6, unlike formalist theories such as T1–T3, gesture toward the physical nature of intelligence, the physical remains an implicit background rather than an active explanatory principle. Defenders of T4 treat the physical as a mere substrate for computation, not a constitutive condition for intelligence as an emergent pattern or property. Defenders of T5 draw attention to the fact that no models of the world are necessary if it can be explored physically. Defenders of T6 focus on the embodied interaction between a system and its surroundings. Both offer minimal specification of how information, energy, and matter gradients are coupled together to yield adaptive and intelligent behaviour (Mengaldo et al., 2022). For convenience, T4, T5 and T6 can be termed tokenist, materialist/embodied theories of intelligence.

One side-effect of top-down, formalist theories of intelligence and tokenist, materialist/embodied theories of intelligence is that the thermody-

dynamic costs of information-processing have remained largely invisible or hidden from view. For instance, state-of-the-art AI systems impose astronomical environmental costs: training a single BERT base model, without hyperparameter tuning, on GPUs has been estimated to require as much energy as a trans-American flight (Bender et al., 2021, p. 612). Following Wittgenstein, we propose returning to the rough ground and re-theorizing intelligence afresh and from the ground up, such that matter, energy, and information gradients can be re-coupled together and intelligence can once again be recognized as physically instantiated, subject to thermodynamic constraints, and best described in terms of information/energy trade-offs. As we shall argue, this new theory of intelligence — which we shall term a Matryoshka model of intelligence — constitutes a necessary step toward the theory-guided design of more sustainable AI systems.

§2 Nested ontology

Our theory of intelligence is best interpreted in the broader context of a nested ontology:

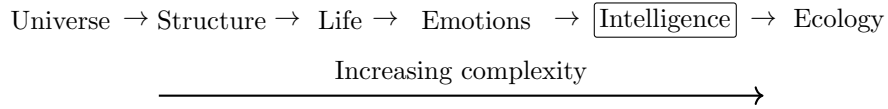


Figure 1: The nested ontology associated with our hierarchical, thermodynamically grounded theory of intelligence

Each node represents a higher level of complexity or structural organization than the previous node, with the higher levels emerging from the lower levels of complexity, while introducing new organizational principles

and constraints. Furthermore, each level of complexity or structural organization is sustained by a specific coupling logic that gives rise to emergent patterns and properties. Intellectual precedents for our approach include big-historical and big-philosophical attempts to make sense of how things in the broadest possible sense of the term hang together in the broadest possible sense of the term; emergentism (philosophy of mind) and the hierarchy of sciences (philosophy of science) in philosophy; complexity theory; the theory of dissipative structures; cybernetics; the theory of prediction error minimization; the theory of connectivity; and the metabolic theory of ecology (Adams, Shipp, and Friston, 2013; Brown et al., 2004; Carhart-Harris and Friston, 2010; Christian, 2004, 2018; Clark, 2013; Clayton and Davies, 2006; Collier, 1996; Comte, 1830–1842; Cunningham, 2001; Friston, 2005, 2009; Friston and Stephan, 2007; Gillooly et al., 2001; Hohwy, 2013; Kauffman, 2008; Ladyman, 2020; Ladyman and Ross, 2007; Laughlin, 2005; Li, Liu, and Tsien, 2016; Meer, 2006; Prigogine and Lefever, 1968; Prigogine and Nicolis, 1967, 1985; Prigogine and Stengers, 2018; Sellars, 1963; Tsien, 2015; West, Brown, and Enquist, 1997; Wiener, 1948/2019; Xie et al., 2016).

§2.1 Universe

At the simplest and most fundamental level, we have the universe or the totality of physical existence, governed by the laws of physics. A featureless, structure-free universe has neither matter, stars, energy fields, nor any curvature of space or time: it is the perfect sameness of pure, flat spacetime.¹ This perfect sameness gives rise to three translational symmetries in space, one translational symmetry in time, three rotational symmetries in space,

¹ We could think of a structure-free universe as analogous to an empty stage with neither actors, props, scenery, nor any preferred spots or directions.

and three boost symmetries in space, or the ten $(3 + 1 + 3 + 3)$ Poincaré symmetries in the Poincaré group (Minkowski, 1908; Poincaré, 1906). These Poincaré symmetries express the general idea that the laws of physics are the same everywhere, at all times, and for all inertial observers.²

§2.2 Structure

According to the theory of dissipative structures, the first structures emerge as a result of symmetry-breaking and open systems maintaining their local order by exchanging matter and energy with their surroundings. These structures are termed ‘dissipative’, not least because the reduction in local entropy ($\Delta S_{\text{system}} < 0$) is coupled to the increase in the entropy of the surroundings ($\Delta S_{\text{surroundings}} > 0$), in accordance with the second law of thermodynamics ($\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$).³

Building upon these foundational dissipative processes, increasingly complex structures arise at multiple nested levels (Fig. 1). At the molecular and cellular level, self-organizing reactions give rise to metabolic networks and organelles that sustain life, store information, and process energy

² By a symmetry is meant invariance under specific transformations. A further theorem, Noether’s (1918/1983) theorem, asserts that each continuous symmetry in nature corresponds to a conservation law. When applied to the Poincaré group, Noether’s theorem implies the conservation of energy, momentum, and angular momentum. All things considered, Noether’s theorem reveals a deep structural unity and elegant coherence in our physical universe and lends credence to the claim that physics is the study of symmetry and symmetry-breaking (Anderson, 1972).

³ Entropy refers to a measure of the number of ways in which a system can be arranged and it is often taken to be a measure of disorder, uncertainty, or randomness. Where S denotes entropy, k_B denotes the Boltzmann constant, $\ln$ denotes the natural logarithm function (log base e), and W denotes the number of possible microstates (specific arrangements of particles) of a system consistent with its macroscopic state (large-scale properties), entropy may be mathematically defined in statistical mechanics as $S = k_B \ln W$ (Boltzmann, 1896; Planck, 1914).

efficiently. At the level of neural systems, structured connectivity and modular organization enable the selective routing of information, learning, and memory formation. These structural motifs optimize the trade-off between energetic cost, computational efficiency, and robustness to perturbations. Beyond individual organisms, higher-order structures — including body plans, organ systems, and social organizations — constrain and shape how agents interact with their environment and with one another. These structures determine the flow of energy, information, and resources, constraining possible behaviours while facilitating adaptability and coordination.

Finally, ecological and environmental structures provide the outermost layer of organization, setting the broader physical and informational context in which intelligence evolves and operates. Resource availability, spatial distribution, and temporal dynamics define the energy and information landscape, ultimately constraining the efficiency and sustainability of adaptive behaviours. Recognizing structure across these nested levels allows us to frame intelligence not merely as abstract computation but as a physically instantiated phenomenon, shaped by energy, matter, and information across varying scales.

§2.3 Life

Following the emergence of structure in an originally structure-free universe, the next complexity threshold to be crossed involves the emergence of life from non-living, physical matter. This is known as abiogenesis. Several researchers have argued that the origin of life on our planet Earth can be traced to the coupling of matter gradients (proton gradients) to energy gradients (Lane, 2015; Martin and Russell, 2007; Russell, Hall, and Martin, 2010; Sojo et al., 2016). More specifically, there is a steep pH or proton gradient

($\Delta\text{pH} = \text{pH}_{\text{vent}} - \text{pH}_{\text{ocean}} \gg 0$) between the alkaline vent fluid ($\text{pH}_{\text{vent}} > 7$) and the acidic ocean water ($\text{pH}_{\text{ocean}} < 7$). This proton gradient, in turn, induces a proton-motive force ($\Delta\mu_{\text{H}}$) from a high-proton concentration (acidic) toward or in the direction of a low-proton concentration (alkaline). The movement of protons, driven by the proton-motive force $\Delta\mu_{\text{H}}$, is an exergonic or energy-releasing process ($\Delta G < 0$) that can drive spontaneous reactions, create organic molecules, and give rise to proto-metabolic networks.⁴ Although local entropy decreases as larger, more complex organic molecules such as nucleic acid molecules and proteins are synthesized from simpler constituents ($\Delta S_{\text{system}} < 0$), the entropy of the universe is still increasing in accordance with the second law ($\Delta S_{\text{universe}} \geq 0$), since at least some energy and matter are being dissipated as heat (useless energy) or chemical by-products (material waste) at these vents ($\Delta S_{\text{surroundings}} > 0$). This eventually gives rise to the first forms of life — archaea — at these vents (DeLong, 1992; Woese and Fox, 1977).

The coupling of matter to energy gradients begins with external proton gradients present across the surfaces of hydrothermal vents. These gradients are eventually harnessed by ancient proton pumps in archaea and bacteria, incorporating them into the internal energy economy of single-celled organisms. In more complex organisms, this coupling of matter to energy gradients extends to mitochondria, where proton gradients across the inner mitochon-

⁴ For any chemical reaction:

$\Delta G = \Delta H + T\Delta S$, where G denotes Gibbs free energy, H denotes the enthalpy of a system (a measure of its total heat content at constant pressure), S denotes its entropy (a measure of its disorder), and T denotes its temperature

If $\Delta G = 0$, a reaction is at equilibrium and there is no net change. If $\Delta G < 0$, a reaction is exergonic (energy-releasing) and can happen on its own or spontaneously without any energy input. If $\Delta G > 0$, a reaction is endergonic (energy-absorbing) and needs an energy input.

drial membranes drive ATP synthesis and support diverse metabolic pathways.⁵ Life as we know it is therefore supported by a metabolic principle, according to which catabolic processes (F_{cat} or catabolic flux, representing the energy released per unit time as nutrients are broken down into their simpler constituents to release energy) are coupled to anabolic processes (F_{ana} or anabolic flux, representing the energy consumed per unit time as more complex molecules such as proteins, DNA, RNA, and tissues are built and energy is stored in chemical bonds) in accordance with the second law. Where ΔE_{system} refers to the net energy change stored in a living organism relying on metabolic pathways, $\Delta E_{\text{system}} = F_{\text{cat}} - F_{\text{ana}}$. At least this much energy will eventually have to be dissipated as heat, implying that $T\Delta S_{\text{surroundings}} \geq \Delta E_{\text{system}}$.⁶

§2.4 Emotions

Between life and intelligence lies an intermediate level of organization: emotions. Within our present framework, emotions are neither subjective phenomena nor purely cognitive states, but emergent regulatory processes grounded in metabolic and neural dynamics. They arise from the interaction of energy throughput, neurochemical modulation, and neural network activity, and operate as low-dimensional control variables that shape behaviour under energetic and informational constraints (Zall et al., 2025).

Emotions function by integrating signals related to internal metabolic

⁵ See also the connection between proton leak in the inner mitochondrial membranes, metabolic entropy, and endothermy in § 3.

⁶ A reversible process is a process in which the system and environment can be restored to exactly the same initial states that they were in before the process occurred. As living organisms are irreversible systems, we must concede that $T\Delta S_{\text{surroundings}} > \Delta E_{\text{system}}$ in practice.

state, environmental conditions, and ongoing activity, thereby providing rapid feedback about the efficiency or inefficiency of current actions. Rather than encoding detailed representations of the world, emotional states compress complex evaluations into simple signals that bias attention, learning, and action selection. In this sense, emotions mediate the transition from metabolic regulation to adaptive behaviour, preparing the ground for higher-level cognitive processes (Mao et al., 2025).

Importantly, both positive and negative emotions play adaptive roles. States associated with satisfaction or reward tend to stabilize ongoing strategies, while states such as frustration, stress, or dissatisfaction signal persistent prediction error or diminishing returns. From a thermodynamic perspective, persisting in such states would imply continued energy expenditure with limited informational gain. Emotional signals therefore act as stopping or switching mechanisms, promoting exploration, strategy revision, or disengagement when necessary. By occupying this intermediate level, emotions couple metabolic constraints to emerging cognitive control. They ensure that intelligence does not operate in isolation from energetic reality, but remains responsive to costs, limits, and opportunities. In the Matryoshka model, emotions thus constitute a necessary layer of organization, linking life to intelligence through physically grounded regulation rather than abstract computation alone.

§2.5 Intelligence

Yet another complexity threshold involves the emergence of intelligence from life: biological intelligence emerged in three of the five animal clades — ctenophores, cnidarians, and bilaterians — and is typically supported by a

nervous system.⁷ Neurons, in particular, are specialized cells that can form neural networks to support the high-speed conduction of signals. Neural networks are metabolically expensive to support: each adult human brain constitutes about 2% of the total body mass but consumes about 20% of the total energy budget (Padamsey and Rochefort, 2023). At the same time, neural networks allow for matter and energy gradients to be coupled to information gradients: at ion pumps across the plasma membranes of neurons, a matter gradient (Na^+ and K^+) is used to generate an energy gradient, and these matter and energy gradients are in turn coupled to an information gradient, resulting in the conversion of electrochemical energy into electrical signals (action potential) for high-speed signal conduction and information-processing.

§2.6 Ecology

Ecology constitutes the outermost level of organization in the Matryoshka model, emerging from the interactions of multiple intelligent systems embedded within shared physical environments. At this level, intelligence is no longer an individual property but a distributed phenomenon arising from networks of agents that exchange matter, energy, and information. Ecological systems are sustained by flows of resources and by the coupling of cooperation and competition, enabling local order to emerge while overall entropy continues to increase.

⁷ At the same time, the biological neural substrate for animal intelligence is multiply realizable: comb jellies and other ctenophores have a continuous neural network with a non-synaptic architecture, echoing Golgi’s (1885) reticular theory, whereas cnidarians and bilaterians have a discrete neural network with a synaptic architecture and neurons as their fundamental units, echoing Cajal’s (1911) neuron doctrine. For more on the thesis of multiple realizability, see Putnam (1967).

From a thermodynamic perspective, ecological organization depends on the circulation and dissipation of energy across populations and environments. Individual agents act as information engines that extract work from local gradients, but ecological stability requires that these activities remain compatible with global resource constraints. As energy and information flow through social and ecological networks, inefficiencies manifest as waste, redundancy, noise, or conflict (informational analogues of heat dissipation). These processes are unavoidable and play a role analogous to entropy production at lower levels of organization.

Ecological dynamics also impose constraints on intelligence itself. The success of an intelligent system depends not only on its internal efficiency but also on its ability to coexist, coordinate, and compete within a broader network of agents. Mechanisms such as communication, social learning, norm formation, and the division of labour allow information to be shared and reused, increasing functional output without a proportional increase in individual energy expenditure. At the same time, ecological pressures can limit unbounded growth in cognitive or computational capacity, reinforcing trade-offs between performance, sustainability, and resilience.

In AI systems, this ecological perspective suggests that intelligence should be evaluated not only at the level of individual models or agents, but also at the level of interacting systems operating within shared computational, energetic, and informational environments. Large-scale AI deployments form information ecosystems with their own forms of congestion, redundancy, and entropy. Designing sustainable AI therefore requires attention to ecological effects, including resource sharing, coordination mechanisms, and the cumulative thermodynamic costs of networked intelligence. Within the Matryoshka model, ecology completes the hierarchy by situating intelligence within the broader dynamics of collective organization and

environmental constraint.

§3 Coupling logic

§3.1 Expansion & collapse at the level of the universe

Across all levels of complexity or structural organization, new patterns or properties emerge from lower-level ones, with each level of complexity or structural organization being sustained by a coupling logic (Table 1). The universe emerges from a coupling together of expansion (driven by dark energy or Λ) and collapse (driven by gravity or G). The second law can be generalized to include the entropy of black holes. It has been demonstrated that black holes have an entropy proportional not to their volume but rather the areas of their event horizon. Furthermore, the entropy of each black hole can be calculated using the Bekenstein–Hawking formula: $S_{\text{BH}} = \frac{4k_B\pi GM^2}{\hbar c}$ (Bekenstein, 1973; Hawking, 1976).⁸ The Bekenstein–Hawking formula implies that a black hole’s entropy increases with its mass. Consequently, the change in cosmic entropy can be approximated as the sum of changes in the entropies of all black holes in the universe ($\Delta S_{\text{cosmic}} \approx \sum_i \Delta S_{\text{BH},i}$).

⁸ k_B denotes the Boltzmann constant, G denotes the gravitational constant, M denotes the mass of the black hole, $\hbar$ denotes the reduced Planck constant, and c denotes the speed of light.

Domain	Type of structural organization	Coupled processes	Mechanisms	Entropy
Cosmology	Universe	Expansion (Λ) $\leftrightarrow$ Collapse (G)	Dark energy (anti-gravity) and gravity	Cosmic entropy ($\Delta S_{\text{cosmic}} = \sum_i \Delta S_{\text{BH},i} + \Delta S_{\text{CMB}} + \Delta S_{\text{matter}} \geq 0$) ^a
Physics	Structure	Symmetry-breaking ($\Delta S_{\text{system}} < 0$) $\leftrightarrow$ symmetry-preserving ($\Delta S_{\text{surroundings}} > 0$)	Fields and forces	Thermodynamic entropy ($\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$)
Chemistry	Reactions	Exergonic ($\Delta G < 0$) $\leftrightarrow$ Endergonic ($\Delta G > 0$)	Chemical bonds and catalysts	Chemical entropy ($\Delta G_{\text{total}} = \Delta G_{\text{exergonic}} + \Delta G_{\text{endergonic}} < 0$) ^b
Biology	Life	Catabolic (F_{cat}) $\leftrightarrow$ Anabolic (F_{ana})	Metabolic pathways	Metabolic entropy ($F_{\text{metabolic}} = F_{\text{cat}} - F_{\text{ana}} \geq 0$) ^c
Psychology	Emotions	Adaptive responses $\leftrightarrow$ Maladaptive responses	Neurochemical modulators (activation and inhibition)	Psychological uncertainty (anxiety)
Neuroscience	Intelligence	Sensory input (μ) $\leftrightarrow$ Motor output (α)	Neural networks (activation, inhibition, and coordination)	Cognitive uncertainty (surprise or prediction error)
Ecology	Networks	Flow of resources (matter, energy, information) $\leftrightarrow$ Energy dissipation	Social networks and populations (competition and cooperation)	Ecosystem entropy

^a $S_{\text{BH},i}$ denotes the sum of changes in the entropies of all i black holes in the universe, S_{CMB} denotes the change in cosmic microwave background radiation entropy, and S_{matter} denotes the change in entropy of baryonic matter

^b For a chemical reaction to proceed spontaneously, it must be the case that $\Delta G_{\text{total}} < 0$. This ensures that the reaction goes forward

^c For the catabolic flux (F_{cat}) to be sufficient to drive the anabolic reactions (F_{ana}), it must be the case that $F_{\text{metabolic}} > 0$

Table 1: A nested ontology with seven hierarchical levels of complexity or structural organization, with each level governed by a distinct coupling logic

§3.2 Symmetry-breaking & symmetry-preserving at the level of structure

Physical structures emerge from a coupling together of symmetry-breaking and symmetry-preserving processes. The emergence of physical structure, in turn, is typically associated with an increase in thermodynamic entropy ($S_{\text{universe}} \geq 0$). Chemical compounds emerge from a coupling together of exergonic and endergonic processes, and the decrease in Gibbs free energy ($\Delta G_{\text{total}} < 0$) drives spontaneous chemical reactions.

§3.3 Catabolic & anabolic processes at the level of life

Living organisms, in turn, rely on a coupling together of catabolic and anabolic processes, and the maintenance of local order in living organisms is typically associated with an increase in metabolic entropy ($F_{\text{metabolic}} \geq 0$). Endothermy offers a vivid manifestation of metabolic entropy: warm-blooded animals such as birds and mammals are able to generate an internal source of heat and internally regulate their body temperature within a high and stable range, primarily because a proton leak across the inner mitochondrial membrane results in a suboptimal ATP yield, with some of the energy that would otherwise have gone into ATP synthesis being dissipated as heat instead (Hulbert and Else, 1999, 2005; Turner et al., 2005).

A high and stable body temperature range facilitates thermoregulation, a healthy rate of enzyme-driven reactions, a high energy throughput, a high ATP turnover, and support for metabolically expensive neurons. One path leads to emotions: neurochemical modulators (dopamine, serotonin, norepinephrine, and so on), important by-products of this metabolic entropy,

mediate emotional states. Another path leads to intelligence: neurochemical modulators influence attention, motivation, arousal, and decision-making, guiding cognition and leading to certain computations being prioritized over others. As a result of a high metabolic throughput, metabolically expensive neurons and complex neural activity can be supported. Complex neural activity generates emotions (mediated by neurochemical modulators) and emotions guide both cognition and behaviour. Behaviour, in turn, feeds back on metabolism and completes the feedback loop. According to the endothermic brain hypothesis, this feedback loop functions essentially as a royal road to the development of large, energetically costly, endothermic brains (Osvath et al., 2024).

§3.4 Emotions as thermodynamic regulators

A central but often underappreciated component of this model of intelligence is the role of emotions as functional, thermodynamically grounded regulators of cognition and behaviour. Emotions are not epiphenomenal or irrational add-ons, but emergent control signals arising from metabolic, neural, and computational dynamics. By modulating attention, motivation, learning rates, and action selection, they efficiently allocate limited energetic, computational, and spatial resources, compressing complex evaluations of internal and external states into low-dimensional signals that can rapidly guide decision-making under uncertainty.

Negative emotions are not failures of intelligence but essential features of adaptive systems. States such as frustration, anxiety, or dissatisfaction correspond to sustained prediction error, poor progress, or inefficient energy-to-information trade-offs. From a thermodynamic perspective, continuing an unproductive course of action wastes energy, computation, and space. Sig-

nals analogous to frustration mark diminishing returns, suppress the current policy, and trigger exploration or strategy switching. In biological systems, such emotions interrupt perseveration and promote behavioural flexibility; in artificial systems, engineered analogues could force a machine to abandon a locally stable but globally inefficient solution, conserving memory, processing power, and hardware resources (Pandelea, Ragusa, Gastaldo, et al., 2023; Pandelea, Ragusa, Young, et al., 2022).

Within our framework, emotions mediate the transition between information processing and action across nested levels of organization. They function as feedback variables linking metabolic cost, cognitive uncertainty, hardware constraints, and behavioural outcomes. An intelligent system without emotional dynamics would lack an intrinsic mechanism for valuing when to persist, when to explore, and when to stop, making it energetically and computationally inefficient. Embedding emotional analogues (even aversive ones) into AI is therefore not about anthropomorphism, but about respecting the physical logic of intelligence: emotions are energy- and hardware-aware control signals that help systems navigate trade-offs, manage entropy, and remain adaptive over time (Marinoni et al., 2025).

§3.5 Sensory input & motor output at the level of intelligence

More generally, intelligence emerges from a coupling together of sensory input (μ) and motor output (α), and can be adequately described in terms of a general model for information-processing behaviour known as the free energy minimization or prediction error minimization model (Adams, Shipp, and Friston, 2013; Carhart-Harris and Friston, 2010; Clark, 2013; Friston,

2005, 2009; Friston, Kilner, and Harrison, 2006; Friston and Stephan, 2007; Hohwy, 2013). According to this model, intelligent organisms aim, when operating behind the veil of a Markov blanket and making predictions based on internal models, to minimize the divergence between actual and predicted sensory inputs — commonly referred to as surprise, prediction error, or uncertainty.⁹ The entropy associated with intelligence, then, is cognitive uncertainty (surprise or prediction error). To minimize surprise or prediction error, intelligent organisms can either update their internal states (including beliefs) to minimize divergence ($\mu = \operatorname{argmin}_{\mu} \text{Divergence}$) or act or intervene on the world to maximize predictive accuracy ($\alpha = \operatorname{argmax}_{\alpha} \text{Accuracy}$) (Fig. 2).

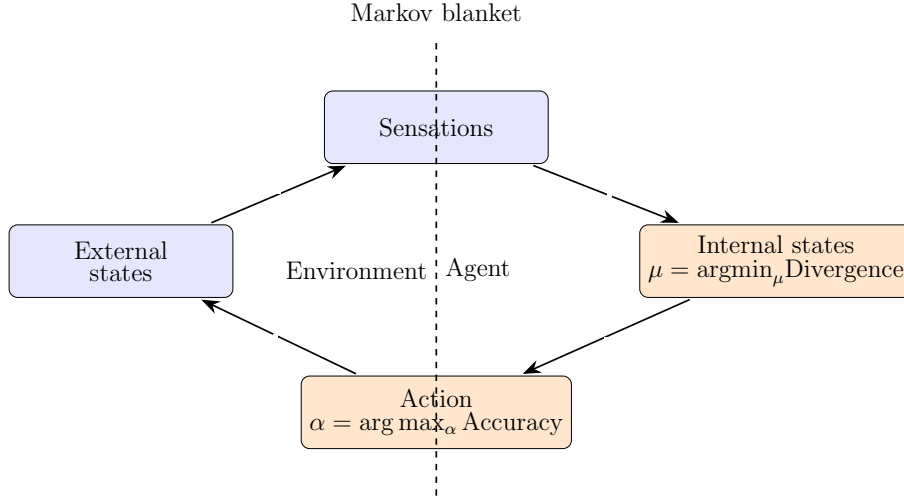


Figure 2: The dynamic interactions between intelligent organisms and their environment, interpreted in terms of the free energy or prediction error minimization account. Adapted from Friston (2009, p. 294)

⁹ Each Markov blanket statistically separates a system’s internal states from its external states, mediating interactions solely via sensory and active states.

§4 Scaling laws

§4.1 The Stefan–Boltzmann law in physics

We have already situated our theory of intelligence within a nested ontology (§ 2). In addition, each level of complexity or structural organization within this nested ontology is characterized in terms of a distinct coupling logic, set of mechanisms, and entropy (§ 3). In this section, we shall argue that this coupling logic can be quantified in terms of scaling laws (typically energy-to-function laws) that connect an energy throughput to a functional output (typically the amount of work performed). In the domain of physics, the Stefan–Boltzmann law connects the temperature of a blackbody T (energy input) to the radiant intensity M or power (energy flux) per unit area (functional output): $M \propto T^4$ (Boltzmann, 1978; Stefan, 1879).¹⁰

§4.2 The law of mass action in chemistry

In the domain of chemistry, the law of mass action in chemical kinetics connects the molar concentrations of reactants A and B (energy input, since the reactants have chemical potential in the form of Gibbs free energy) to the reaction rate r (functional output): $r \propto [A]^m[B]^n$, where the exponents m and n are partial orders of reactions and not necessarily the stoichiometric coefficients (Waage and Guldberg, 1864).

¹⁰ A blackbody is an idealized physical body that absorbs all incident electromagnetic radiation, does not reflect or transmit any radiation, and emits radiation according to its temperature alone.

§4.3 Quarter-power scaling laws in biology

In the domain of biology, the metabolic theory of ecology identifies the metabolic rate as the key variable controlling biological and ecological processes (Brown et al., 2004; Gillooly et al., 2001; Meer, 2006; West, Brown, and Enquist, 1997). The basal metabolic rate, representing the minimum energy expenditure required for a living organism to maintain certain vital functions (breathing, blood circulation, cellular metabolism, and so on) at rest, provides living organisms with an energy throughput and an energy input. A host of scaling laws connects the basal metabolic rate B (energy input) to a range of functional outputs:¹¹

$M \propto B^{\frac{4}{3}}$, where M denotes the body mass of an organism (also known as Kleiber's (1947) law);

$A \propto B$, where A denotes the cross-sectional areas of tree trunks;

$f \propto B^{-\frac{1}{3}}$, where f denotes the heart rate;

$r_{\max} \propto B^{-\frac{1}{3}}$, where $r_{\max}$ denotes the maximum population growth rate;

$T \propto B^{\frac{1}{3}}$, where T denotes an organism's lifespan;

$t_c \propto B^{\frac{1}{3}}$, where t_c denotes the blood circulation time.

§4.4 Energy-to-neuron scaling law in neuroscience

In the domain of neuroscience, a scaling law connects the brain's energy consumption E (measured as glucose metabolism) to the total number of

¹¹ These scaling laws are originally allometric or body mass-related, but have been adjusted, for argumentative purposes, to an energy-to-function form. To derive the original allometric form of these laws, we simply need to replace B with M (body mass) and adjust the exponent values using the $M \propto B^{\frac{4}{3}}$ relation.

neurons N supporting brain function: $E \propto N$ (Herculano-Houzel, 2011). This indicates that, across species, each neuron receives a roughly fixed energy budget, with important implications for both emotional complexity (supported by neurochemical modulators) and cognitive processing (guided by emotions and supported by neural networks). More broadly, metabolic constraints shape the capacity for neural computation and the modulation of neurochemical signals underlying emotions. In addition, there are two connectivity-related scaling laws in the theory of connectivity (Li, Liu, and Tsien, 2016; Tsien, 2015; Xie et al., 2016):

$N_{\text{cliques}} = 2^i - 1$, where N_{cliques} denotes the number of neural cliques with all possible combinatorial connectivity patterns and i denotes the number of distinct information inputs;
 $Pr = \frac{2^{i-1}}{b}$, where Pr denotes the number of presynaptic neural cliques required to cover postsynaptic convergence and b denotes the number of axon branches.

Each neural clique — also termed a cell assembly — is a group of neurons that can fire transiently or sequentially. Through these firing patterns, neural cliques can represent percepts, memories, and concepts. It has even been argued that these neural cliques or cell assemblies can be regarded as the working units of the nervous system (Hebb, 2005; Semon, 1911). Suppose that we rewrite Houzel’s (2011) energy-to-neuron scaling law (pertaining to individual neurons) as $E \propto N_{\text{neurons}}$. Furthermore, suppose that each neural clique is made of a certain number of neurons $n_{\text{per clique}}$, such that $N_{\text{neurons}} = N_{\text{cliques}} \times n_{\text{per clique}}$. If we assume that $n_{\text{per clique}}$ is roughly constant, then we can derive $E \propto N_{\text{cliques}} = 2^i - 1$. Furthermore, since $Pr = \frac{2^{i-1}}{b}$, $2^{i-1} = b \times Pr$. Consequently, $2^i = 2b \times Pr$ and $N_{\text{cliques}} = (2b \times Pr) - 1$. If $(2b \times Pr) \gg 1$, then we can approximate N_{cliques} as $N_{\text{cliques}} \approx 2b \times Pr$.

Therefore, $E \propto N_{\text{neurons}} = N_{\text{cliques}} \times n_{\text{per clique}} \approx (2b \times Pr) \times n_{\text{per clique}}$. This formula tells us is that the energy consumption of the brain E scales linearly with presynaptic connectivity Pr up to a factor of $2b \times n_{\text{per clique}}$. As postsynaptic neural cliques can sum across inputs from presynaptic neural cliques, postsynaptic cliques effectively implement a power-of-two-based combinatorial code when represent combinations of presynaptic inputs. More distinct percepts, memories, and concepts can be represented by either increasing the connectivity per presynaptic clique ($\uparrow$ in b), increasing the number of presynaptic cliques ($\uparrow$ in Pr), or increasing the size of each clique ($\uparrow$ in $n_{\text{per clique}}$). The combinatorial coding in postsynaptic cliques, in turn, allows for energy-efficient information-processing: each neural network can represent far more patterns (percepts, memories, and concepts) than the total number of neurons (N_{neurons}) it possesses, although the energy budget of this network scales linearly with N_{neurons} .

§5 Engines

Biological intelligences are more than merely emergent patterns or properties at a certain level of description or structural organization (for instance, when the metabolic pathways are sufficiently sophisticated to support the metabolic costs of brain tissue, nervous systems, and neural networks). Biological networks can also be regarded as engines. A system is an engine, just in case it can extract useful work from gradients (matter, energy, or information), while adhering to the second law of thermodynamics.

A Carnot engine refers to a heat engine that can extract work from a thermal gradient between a hot reservoir (at temperature T_H) and a cold reservoir (at temperature T_C). The maximal thermal efficiency of a Carnot

Domain	Emergent pattern or property	Energy input	Functional output	Scaling law
Physics	Blackbody radiation	Temperature T	Power per unit area M	$M \propto T^4$
Chemistry	Chemical reactions	Gibbs free energy G of reactants A and B	Chemical reaction rate (r)	$r \propto [A]^m[B]^n$
Biology	Life-essential processes	Catabolic energy throughput from basal metabolic rate (B)	Body mass (M), cross-sectional area of tree trunks (A), heart rate (f), maximum population growth rate ($r_{\max}$), lifespan (T), blood circulation time (t_c)	$M \propto B^{\frac{4}{3}}; A \propto B; f \propto B^{-\frac{1}{3}}; r_{\max} \propto B^{-\frac{1}{3}}; T \propto B^{\frac{1}{3}}; t_c \propto B^{\frac{1}{3}}$
Neuroscience	Intelligence and information-processing	Energy consumed by the brain E	Total number of neurons supporting brain function N	$E \propto N$

Table 2: The scaling laws within the nested ontology

engine can be computed as $\eta = \frac{\text{Work done by the engine per cycle}}{\text{Heat absorbed per cycle}} = 1 - \frac{Q_C}{Q_H}$, where Q_H denotes the heat absorbed from the hot reservoir and Q_C denotes the heat released to the cold reservoir.¹² If any engine were more efficient than a Carnot engine, then it would release less or no heat to the cold reservoir ($Q_C \rightarrow 0$), resulting in a net decrease in the entropy of the universe and a violation of the second law ($\Delta S_{\text{universe}} < 0$).

Maxwell's (1871) demon, by contrast, refers to a single-reservoir system relative to which an intelligent being (termed a 'demon') performs a molecule-sorting exercise. This single reservoir is filled with a gas that initially has a uniform temperature. A door partitions this single reservoir into two chambers: a left chamber and a right chamber. By controlling this door, Maxwell's demon selectively lets faster-moving gas molecules go to the right chamber and slower-moving gas molecules go to the left chamber. After this molecule-sorting exercise has been completed, a thermal gradient will be generated. The molecule-sorting exercise of Maxwell's demon involves measuring the relative velocities of gas molecules, storing this information in the memory, and resetting (memory erasure), and measurement, information storage, and memory erasure all have thermodynamic costs. Through this meticulous, demon-driven molecule-sorting exercise, the gas entropy is reduced by $\Delta S_{\text{gas}} < 0$. The second law implies that at least $\Delta S > |\Delta S_{\text{gas}}|$ must be produced elsewhere to ensure that $\Delta S_{\text{universe}}$ does not decrease. According to the Landauer-Bennett thesis, erasing n bits of memory at temperature T_{env} will cost at least $Q_{\text{erase}} \geq k_B T_{\text{env}} (\ln 2) n$ (Bennett, 1982, 2003; Landauer, 1961, 1991). Therefore, if Maxwell's demon must erase information to operate in a new cycle, the heat that is dissipated from memory erasure (ΔS_{erase}) and other information-processing processes cancels out the reduction in gas

¹² η tells us what fraction of the heat absorbed is being converted into useful work per Carnot cycle.

entropy ($\Delta S_{\text{gas}} < 0$) from the molecule-sorting exercise.

The Carnot engine is a heat engine: it extracts useful work from a thermal gradient between two reservoirs at different temperatures. By contrast, Maxwell’s demon is an information engine: it extracts useful work from an information gradient. Both engines incur thermodynamic costs as work is performed, since at least some energy is inevitably dissipated as heat ($Q_C \gg 0$), ensuring that the second law is preserved. Were the intelligence of Maxwell’s demon a biological one, then we would be able to say, in equal measure, that it is sustained by metabolic means and involves a cascade of matter, energy, and information gradients.¹³ All things considered, intelligence, as manifested in information engines, does not violate the second law but rather constitutes a special case of it, implemented through metabolic organization and the cascade of matter, energy, and information gradients.

§6 A Matryoshka model of intelligence

Our analysis supports the notion that a theory of intelligence should be embedded within a nested ontology (§ 2). This nested ontology, in turn, can be construed in terms of a set of Matryoshka or Russian tea dolls of decreasing

¹³ The neural networks of biological intelligences do not run on abstract information alone: the information-processing machinery must be physically instantiated (for instance, in neurons, enzymes, and so on) and maintained through a metabolic energy flow or throughput that is derived from chemical energy. Like all other information engines, biological intelligence must be embodied, energetically grounded, and thermodynamically constrained. A series of nested and interdependent matter (differing Na^+ and K^+ concentrations at ion pumps across the plasma membranes of neurons), energy (chemical potential energy stored in ATP), and information (sensory input) gradients must be coupled together in the appropriate manner: metabolic energy flows sustain the physical structure that enables thermodynamically-constrained information-processing, which in turn regulates the flow of matter and energy in a feedback loop (§ 3).

size, with one doll placed inside another. Each level of this nested ontology is represented by a Matryoshka doll of a particular size, and it corresponds to a particular level of complexity or structural organization. Dolls nested within this doll represent lower levels of complexity, whereas the dolls within which this doll is nested represent higher levels of complexity. Each level of complexity is characterized by the coupling together of complementary processes to maintain local order at that level (§ 3). For convenience, we shall term our theory of intelligence a Matryoshka model of intelligence.

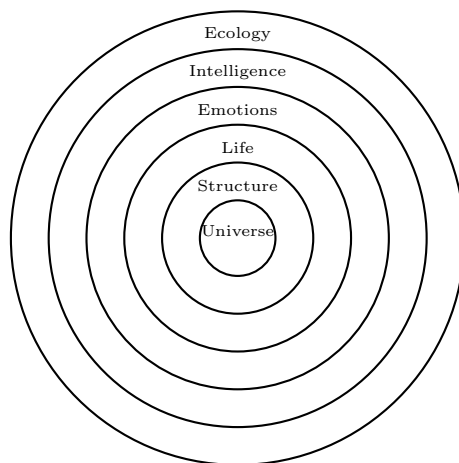


Figure 3: A Matryoshka model of intelligence

The mind-body problem, traditionally regarded as the most central problem in the philosophy of mind, is concerned with the nature of the relationship between the mind (thought, intelligence, consciousness) and the body (matter) (McLaughlin, Beckermann, and Walter, 2009; Szubka and Warner, 1994). Since the smallest doll in this set of Matryoshka dolls corresponds to the universe (the simplest and most fundamental level of structural

organization), our Matryoshka model implies a materialism about the mind. A distinct response to the mind-body problem, materialism asserts that everything that exists is made of one fundamental substance: physical matter. More generally, everything that exists is made of matter, energy, and information flows.¹⁴

Our Matryoshka model further implies that intelligence must be physically instantiated. After all, intelligence is an emergent pattern or property of certain types of organized matter (for instance, living organisms) that can access energy flows to process information in time. Intelligence can be multiply realized in diverse physical instantiations, whether in the form of the non-synaptic neural architectures of comb jellies, the diffuse neural nets of jellyfish, the brains and nervous systems of bilaterians, or even the special case of octopuses having both a central brain and eight arms functioning as semi-independent nervous systems (Sumbre et al., 2001). An abstract mind, shorn of a physical substrate — including neurons, circuits, brains, transistors, photons, or any other medium capable of storing, transmitting, and processing information — would be causally inert. Without physical instantiation, a mind would be bereft of support from any physical carriers of information, making information-processing impossible. Therefore, our Matryoshka model supports Landauer’s dictum: information is and must be physical.

Our Matryoshka model concedes the physical and thermodynamic limits of intelligence from the outset. Although intelligence can be multiply realized, whether in carbon-based biological intelligences inhabiting our planet Earth, non-carbon-based AI systems that have the appropriate level of sophistication and complexity, or even alien life forms, all physical instanti-

¹⁴ Philosophers will recognize this view as physicalism, according to which everything that exists supervenes on the physical (Hellman and Thompson, 1977; Poland, 1994).

ations of intelligence must still satisfy the constraints of space, time, and energy, and the logic of evolutionary history. Paul Thagard (2002, 2025) has defended a similar view that he terms coherent materialism. Information engines, however, sophisticated or intelligent, cannot violate the second law and must operate within the Carnot limit (§ 5). Therefore, although biological intelligences can rely on metabolic means to exploit cascades of matter, energy, and information gradients, process information, and increase the amount of work extracted per unit energy, cognition and computation necessarily incur thermodynamic costs. This suggests that we could measure intelligence in terms of informational efficiency (η_{info}) or a trade-off between information gain and energetic cost.

$\eta = \frac{\text{Work done by the engine per cycle}}{\text{Heat absorbed per cycle}} = \frac{\text{Work}_{\text{useful}}}{Q_H} \leq 1$ allows us to compute the thermal efficiency of a heat engine. By analogy, the informational efficiency of an information engine can be computed as $\eta_{\text{info}} = \frac{I_{\text{useful}}}{E_{\text{consumed}}/k_B T \ln 2} = \frac{I_{\text{useful}} k_B T \ln 2}{E_{\text{consumed}}} \leq 1$. Where Q_H and E_{consumed} respectively represent the energy inputs and $\text{Work}_{\text{useful}}$ and I_{useful} the functional outputs of heat and information engines, it might be equally profitable for us to think in terms of scaling laws or energy-to-function laws that connect an energy throughput to a functional output (§ 4). Since there is an upper bound to efficiency (thermal or informational), we can understand why the energy input must scale with any increase in functional capacities (including information-processing capacities). The second law provides us with the physical basis for a host of no-free-lunch theorems: any increase in mechanical work (physics), chemical reaction rates (chemistry), life-essential and metabolic functions (biology), emotional sophistication (psychology), cognition (neuroscience), or related functional capacities necessarily entails a proportional or greater increase in energy input. At the same time, any increase in connectivity allows for more energy-efficient information-processing, paving the way for information-sharing in

social networks and information ecosystems at a higher level of complexity or structural organization (ecology). However, information ecosystems also have their associated entropy, whether in the form of noise, misinformation, disinformation, redundancy, or the attention economy. These are the informational equivalents of heat (dissipated energy), representing the inevitable dissipation that follows the generation and maintenance of order.

§7 Implications for AI research

The Matryoshka model provides a new perspective for AI research by treating intelligence as a physically grounded process rather than a purely abstract computation. This perspective has several implications. First, it highlights the importance of considering energy and resource constraints in AI system design: efficient algorithms and architectures should minimize unnecessary computational and energetic costs while maximizing adaptive performance. Second, it encourages hierarchical and modular architectures that mirror the nested coupling observed in biological systems, where lower-level sensory and motor processes interact with higher-level cognitive and decision-making processes. Third, it motivates the integration of feedback mechanisms and dynamic adaptation, ensuring that artificial systems can respond to changing environmental conditions efficiently and robustly. Finally, grounding AI in physical and thermodynamic principles opens the door to developing more sustainable and resilient systems, capable of long-term operation without excessive resource consumption, and better aligned with biological models of intelligence.¹⁵

¹⁵ We agree that lower energy consumption on its own does not necessarily imply higher intelligence. A system may consume minimal energy and yet exhibit little or no adaptive or information-processing abilities. Our central claim rather is that intelligence

§8 Conclusion

Our Matryoshka model reframes intelligence as a physically instantiated phenomenon arising from the coupling of energy, matter, and information across nested levels of complexity. By integrating principles from thermodynamics, metabolism, and information theory, this model provides a unified account of how intelligent behaviour emerges from basic physical and biological processes. Emotions, as by-products of metabolic and neural processes, naturally arise in this framework and serve adaptive functions, helping systems respond to uncertainty and guide decision-making (even negative emotions can signal the need to change strategy). For AI research, our model underscores the importance of considering energetic and material constraints, hierarchical organization, and adaptive feedback mechanisms when designing computational systems. Such an approach not only promotes more efficient and sustainable AI architectures but also aligns artificial systems more closely with the principles underlying natural intelligence. By recognizing intelligence as a product of physical instantiation and metabolic dynamics, with emotions playing a functional role, the Matryoshka model offers both a theoretical foundation and practical guidance for next-generation intelligent systems.

is physically instantiated and therefore subject to thermodynamic constraints. This introduces certain trade-offs between energy expenditure, information processing, and adaptive performance. What matters, then, is not the minimization of energy use but rather energy-information efficiency given various task- and environment-related constraints. Indeed, our perspective justifies an appeal to biological forms of computing, since they typically operate in domains where informational gain per unit of energy consumed is relatively high.

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